Research Notes: Aligning Multimodal Earnings Call Sequences with High-Frequency Market Returns

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1 Introduction and Research Motivation

Earnings conference calls (ECCs) are the primary channel through which public companies disclose forward-looking information to capital markets. Each call produces two jointly available and naturally aligned modalities: a written transcript and an audio recording. Prior work has established that both what managers say (the semantic content of the transcript) and how they say it (acoustic features such as pitch variability, speaking rate, and disfluency) carry incremental information for predicting post-call abnormal returns and realized volatility [3, 6, 1, 7].

The key data constraint motivating this paper is not text-to-audio alignment—that problem is solved (partially with 80-90% accuracy?), because the transcript and the audio recording can be paired at the utterance level using standard forced-alignment tools. The unresolved alignment gap lies between the multimodal utterance sequence and the concurrent high-frequency (HF) return series. Transcript data do not carry reliable within-call clock times. A vendor-supplied ECC transcript reports utterances in sequential order, but the precise calendar time at which each sentence was spoken is generally not recorded. High-frequency market data, available at the tick or minute level, therefore cannot be matched to specific utterances without knowing when each utterance occurred.

This creates a well-posed machine learning problem. We have two ordered sequences: (i) N multimodal utterance representations covering the content of the call, and (ii) T minute-level observations of the stock's abnormal return and trading volume covering the same call window. Both sequences share a known time axis over the call duration $[\tau_s, \tau_e]$, but the position of each utterance on that axis is latent. Recovering or approximating these latent positions—what we call the *ECC-to-HF alignment problem*—is the methodological focus of this paper.

Solving this problem matters for two reasons. First, it enables sequence-level prediction of the intraday abnormal-volatility profile as a function of call content, which is strictly more informative than predicting a single post-call scalar from a bag of utterances. Second, if acoustic features improve alignment quality over text alone, this provides direct causal evidence that vocal delivery carries timing-relevant information: hesitation and elevated vocal energy co-occur with the market reactions they help produce, not just in aggregate but utterance by utterance.

This paper targets the ACM International Conference on AI in Finance (ICAIF) as the primary venue, with an AAAI main-track framing available if the Soft-DTW latent-timestamp method is developed into a general-purpose weakly-supervised alignment benchmark.

2 Related Work

2.1 Multimodal ECC Analysis

The MAEC dataset [2] established the first large-scale benchmark for multimodal ECC research, pairing audio and transcripts from over 4,400 calls with stock price labels. Its authors explicitly note that fine-grained alignment within the call can be inaccurate due to audio quality degradation, which directly motivates the weak-alignment framing adopted here. Qin and Yang [3] demonstrated at ACL 2019 that verbal content and vocal cues (pitch, energy, speaking rate) jointly predict post-call realized volatility, validating the multimodal premise. Ye et al. [4] showed at IJCAI 2020 that multi-round Q&A structure carries specific predictive information beyond the prepared remarks, supporting the Q&A focus in this paper. The VolTAGE model [5] applied temporal-aware graph embeddings to ECC sequences for volatility prediction, but still relies on call-level aggregation rather than within-call alignment.

2.2 Acoustic and Vocal Cue Research in Finance

Mayew and Venkatachalam [6] were among the first to show in the *Journal of Finance* that managerial affective states, as measured by vocal analysis, predict future firm performance. Baik et al. [7] recently published in the *Journal of Accounting and Economics* that vocal delivery quality in ECCs predicts earnings-related outcomes above and beyond the text. Pope [1] documents persistent market under-reaction to vocal delivery signals, suggesting that traders do not fully incorporate acoustic information at the time of the call.

2.3 Weakly Supervised and Unaligned Multimodal Learning

The Multimodal Transformer of Tsai et al. [8] is the seminal work on aligning asynchronous multimodal sequences using cross-modal attention, directly relevant to our latent-clock-position framing. D^3TW [9] introduces discriminative differentiable dynamic time warping for weakly supervised action alignment in video, which is structurally the closest ML precedent to the latent-timestamp approach in Strategy D. Cuturi's Sinkhorn distances [15] and Soft-DTW [16] provide the algorithmic foundations for Strategies C and D respectively.

2.4 Text-Driven Volatility and Market Microstructure

Kogan et al. [10] demonstrated at NAACL 2009 that text features from financial reports reliably predict post-announcement realized volatility, establishing the theoretical link between linguistic content and second-moment price dynamics that this paper extends to the within-call setting. Easley et al. [11] showed that abnormal trading volume is a proxy for information arrival in the microstructure literature, providing the theoretical justification for augmenting price-based volatility targets with volume.

3 Problem Formulation

3.1 Data Setting

For each ECC, three artifacts are available.

Artifact 1: Multimodal utterance sequence. A sequence of N fused text-audio representations $\mathcal{U} = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_N\}$, where each $\mathbf{u}_i \in \mathbb{R}^d$ is the output of the multimodal encoder described in Section 4. The transcript and audio are already aligned at the utterance level, so this encoding is clean.

Artifact 2: Call anchor times. The call start time τ_s and call end time τ_e, recoverable from exchange filings, IR event calendars, or vendor metadata. These define the time window over which the alignment problem must be solved.

Artifact 3: High-frequency return and volume data. A sequence of T minute-level observations $\{(r_t, V_t)\}_{t=1}^{T}$ covering the interval $[\tau_s, \tau_e + \Delta]$ for a post-call window Δ. Here r_t denotes the market-adjusted log-return (stock log-return minus concurrent SPY ETF log-return) and V_t denotes the trading volume in shares at minute t. Market-adjustment is essential: raw returns at the minute level are dominated by market-wide noise, and subtracting the SPY return isolates the idiosyncratic component driven by call content [10].

The alignment gap. What is *not* available is the within-call clock time τ_i of each utterance i. The content sequence $\mathcal{U}$ and the HF observation sequence share the same time axis over $[\tau_s, \tau_e]$, but the position of each $\mathbf{u}_i$ on that axis is latent. A more general problem is the alignment of three sequences because audio and text are not perfectly aligned.

3.2 Core Research Questions

RQ1. Alignment value: Which alignment strategies best recover the latent temporal positions of ECC utterances, measured by improvement in intraday abnormal-volatility prediction relative to bag-of-utterances baselines?

RQ2. Modality value: Does multimodal content (text plus audio) produce better alignment with the HF series than text alone? If so, this implies acoustic features carry *timing-relevant* information, not merely call-level predictive value.

3.3 Prediction Targets

We define three prediction targets that exploit the HF data at different granularities, ranging from a full intraday sequence (T1) to call-level scalars (T2, T3).

T1: Volume-weighted minute-level abnormal volatility sequence. The sequence target is $\mathbf{v} = (v_1, \ldots, v_T)$, where each element is defined as:

$$v_t = r_t^2 \cdot \log(1 + V_t), \tag{1}$$

and r_t is the market-adjusted minute return. The $\log(1+V_t)$ weight follows from the microstructure intuition of Easley et al. [11]: when a CEO says something ambiguous, the price may not move directionally, but algorithmic traders will rebalance aggressively, producing a volume spike. Pure squared returns at the minute level are dominated by bid-ask bounce; weighting by volume anchors the target to genuine information-driven activity. A model with utterance-level alignment can predict this sequence; a bag-of-utterances baseline cannot.

T2: Post-call abnormal aggregate volatility. The scalar $V_{\text{post}} = \sum_{t=T_e}^{T_e+12} v_t$ (sum over the 60 minutes following call end, using the volume-weighted definition from Eq. 1). This target is compatible with both aligned and unaligned models and serves as the common evaluation currency across all baselines.

T3: Coarse within-call abnormal reaction profile. Rather than predicting the full T-length sequence for T1, this target summarises the within-call reaction as a K-dimensional profile vector: $\mathbf{c} = (c_1, \ldots, c_K)$, where $c_k = \sum_{t \in \text{segment}_k} v_t$ for $K = 3$ call segments (opening, prepared remarks, Q&A). This is feasible even with only segment-level alignment (Strategy B) and tests whether call structure systematically shapes the intraday reaction pattern.

4 Multimodal Utterance Encoder

4.1 Text Representation

Each utterance i is encoded by passing its text through FinBERT [12], a BERT variant fine-tuned on financial corpora. The `[CLS]` token embedding is used, yielding $\mathbf{t}_i \in \mathbb{R}^{768}$. All FinBERT parameters are frozen. This is efficient and avoids overfitting on a small ECC dataset.

Implementation note for students. Install HuggingFace Transformers and load FinBERT with `AutoModel.from_pretrained("ProsusAI/finbert")`. Pass each utterance string through the tokenizer (max 128 tokens), run the model with `torch.no_grad()`, and extract the `[CLS]` output. Batch all N utterances in one forward pass per call, then save the resulting $(N, 768)$ tensor to disk. This step is the most compute-intensive; pre-compute and cache all calls before any training loop.

4.2 Audio Representation

Audio features combine two complementary sources. (a) Frozen Wav2Vec 2.0 [13] embeddings: the audio segment corresponding to utterance i is passed through the base Wav2Vec 2.0 model with frozen weights; the last-layer hidden states are mean-pooled to yield $\mathbf{a}_i^{\text{w2v}} \in \mathbb{R}^{768}$. (b) eGeMAPS summary statistics [14]: OpenSMILE extracts 88 interpretable prosodic and spectral features per segment, yielding $\mathbf{a}_i^{\text{ema}} \in \mathbb{R}^{88}$. These are concatenated and projected: $\mathbf{a}_i = \mathbf{W}_a[\mathbf{a}_i^{\text{w2v}}; \mathbf{a}_i^{\text{ema}}] \in \mathbb{R}^{256}$, where $\mathbf{W}_a$ is trained.

Implementation note for students. For Wav2Vec 2.0, use `facebook/wav2vec2-base` from HuggingFace. Load the audio segment for each utterance (cut from the full call audio using PyDub or librosa at the utterance boundary from the transcript), pass through the `Wav2Vec2Model` with `output_hidden_states=True`, and mean-pool the last hidden state. For eGeMAPS, install OpenSMILE (`pip install opensmile`) and call `smileapi.process_signal()` with the `eGeMAPSv02` feature set. Each segment should produce one 88-dimensional row. Both representations should be pre-computed and cached.

4.3 Cross-Modal Attention Fusion

Naive concatenation followed by an MLP tends to let the high-dimensional text modality ($\mathbb{R}^{768}$) overpower the audio ($\mathbb{R}^{256}$) because the gradient norms are unbalanced. Instead, we use a cross-modal attention mechanism in which the acoustic features *query* the text representation, explicitly modelling how a vocal signal (hesitation, elevated energy) modifies the interpretive weight of the concurrent text:

$$\mathbf{u}_i = \mathrm{LayerNorm}\Big(\mathbf{t}_i + \mathrm{MultiHeadAttn}\big(\mathbf{Q} = \mathbf{a}_i W_Q,\ \mathbf{K} = \mathbf{t}_i W_K,\ \mathbf{V} = \mathbf{t}_i W_V\big)\Big), \tag{2}$$

where $W_Q \in \mathbb{R}^{256\times d_{\text{attn}}}$, $W_K, W_V \in \mathbb{R}^{768\times d_{\text{attn}}}$, and $d_{\text{attn}} = 64$ per head with 4 heads. The residual connection $\mathbf{t}_i + (\cdot)$ preserves the full semantic content of the text; the attention term allows audio to re-weight which dimensions of the text representation are most salient given the observed vocal delivery. The final $\mathbf{u}_i \in \mathbb{R}^{768}$ is projected to $\mathbb{R}^d$ ($d = 256$) by a two-layer MLP.

Implementation note for students. Implement Eq. 2 using `torch.nn.MultiheadAttention(embed_dim=64*4, num_heads=4)`. The query is formed by projecting $\mathbf{a}_i$ (dimension 256) to dimension $d_{\text{attn}} \times$ num_heads; keys and values are formed by projecting $\mathbf{t}_i$ (dimension 768). Because the query is a single vector per utterance (not a sequence), pass it as a sequence of length 1 to the attention module: `query.unsqueeze(0), key.unsqueeze(0), value.unsqueeze(0)`. The output will be shape $(1, \text{batch}, d_{\text{attn}})$; squeeze and apply LayerNorm. The whole encoder processes N utterances in batch; the attention is applied *within* each utterance, not across utterances.

5 Alignment Strategies

All four strategies respect the **monotonicity constraint**: utterance i must be assigned to a time no earlier than utterance $i-1$. This encodes the obvious prior that speech is delivered in order. Each strategy produces a soft assignment matrix $\mathbf{P} \in \mathbb{R}^{N\times T}$ with P_{it} giving the weight with which utterance i contributes to the representation at minute t.

5.1 Strategy A: Uniform Prior

The simplest baseline assumes utterances are delivered at a constant rate:

$$t_i = \tau_s + (i-1)\cdot\frac{\tau_e - \tau_s}{N-1}, \quad i = 1, \ldots, N. \tag{3}$$

The assignment matrix is a hard diagonal: $P_{it} = 1$ if $t = t_i$ (rounded to the nearest minute), zero otherwise. This is deterministic, requires no training, and is easily reproduced.

Implementation. Given the call duration in minutes $T_{\text{call}} = (\tau_e - \tau_s)$ and N utterances, create a mapping array `assign = np.linspace(0, T_call-1, N).astype(int)` (integer minute indices). Build the sparse assignment matrix with `scipy.sparse.csr_matrix`. This runs in milliseconds per call.

5.2 Strategy B: Segment-Level Alignment

The call is divided into $K = 3$ segments—preamble/opening, prepared remarks, Q&A—with boundary times estimated from transcript structure. The Q&A boundary is identified as the first speaker turn that matches an analyst name (a simple regex over the transcript diarisation). Within each segment, utterances are mean-pooled into a single segment-level vector; the corresponding HF returns are likewise summed. This reduces the alignment problem to three anchor points, all estimable to within a few minutes, and directly yields the T3 target.

Implementation. Parse the transcript to extract speaker labels. Flag the first utterance where the speaker is not an executive (i.e., matches the word "analyst", "operator", or a known bank name) as the Q&A start. Record its position index n_{qa}. Segment 1: utterances $[1, n_{\text{preamble}}]$; Segment 2: utterances $[n_{\text{preamble}} + 1, n_{\text{qa}} - 1]$; Segment 3: utterances $[n_{\text{qa}}, N]$. Pool each segment separately: `seg_repr = utterances[start:end].mean(0)`. Map each segment to its time window using the known call start/end and estimated Q&A start.

5.3 Strategy C: Content-Reactive Soft Warping via Sinkhorn Optimal Transport

Strategy C learns a soft alignment by solving a regularised optimal transport problem. The key innovation is a bilinear cost matrix $\mathbf{S}$ that scores how well each utterance–minute pair explains the co-occurring market reaction.

Constructing the cost matrix S. Let $\mathbf{h}_t \in \mathbb{R}^d$ be the hidden state of a lightweight TCN or GRU encoder applied to the sliding-window HF feature vector (r_t, V_t, v_t) at minute t (trained jointly). The bilinear similarity between utterance i and minute t is:

$$S_{it} = \sigma\big(\mathbf{u}_i^\top \mathbf{W}_s \, \mathbf{h}_t\big), \tag{4}$$

where $\mathbf{W}_s \in \mathbb{R}^{d \times d}$ is a learned bilinear projection matrix and σ is the sigmoid function. Intuitively, S_{it} is high when the semantic-acoustic content of utterance i is predictive of the HF activity at minute t.

Sinkhorn optimal transport. The soft assignment matrix is:

$$\mathbf{P}^* = \arg\min_{\mathbf{P} \in \Pi} \big\langle -\mathbf{S} + \lambda \, \mathbf{D}, \; \mathbf{P} \big\rangle + \varepsilon \sum_{i,t} P_{it} \log P_{it}, \tag{5}$$

where $\Pi = \{\mathbf{P} \geq 0 : \mathbf{P}\mathbf{1} = \mathbf{1}\}$ (row-stochastic), $\mathbf{D}_{it} = |i/N - t/T|$ is the temporal position discrepancy penalty enforcing monotonicity, λ controls the strength of the ordering prior, and ε is the entropic regularisation. The Sinkhorn–Knopp algorithm solves Eq. 5 by alternately normalising rows and columns of $\exp((\mathbf{S} - \lambda\mathbf{D})/\varepsilon)$, which converges in 20–50 iterations and is fully differentiable. Both λ and ε are learnable scalar parameters.

Implementation note for students. The Sinkhorn iteration is straightforward to implement in PyTorch. Start with the log-domain kernel $\mathbf{M} = (\mathbf{S} - \lambda\mathbf{D})/\varepsilon$ (shape $N \times T$). Initialise $\mathbf{u} = \mathbf{0}_N$ (log-domain row scaling). Iterate for 30 steps: (1) $\mathbf{v} = -\log \sum_i \exp(\mathbf{M} + \mathbf{u}[:, \text{None}])$ (column normalisation in log domain); (2) $\mathbf{u} = -\log \sum_t \exp(\mathbf{M} + \mathbf{v}[\text{None}, :])$ (row normalisation). The assignment matrix is $\mathbf{P} = \exp(\mathbf{M} + \mathbf{u}[:, \text{None}] + \mathbf{v}[\text{None}, :])$. All operations are differentiable; gradients flow back to $\mathbf{W}_s$, λ, and ε. An open-source reference implementation is available in the `geomloss` library (PyPI: `pip install geomloss`).

For the HF encoder $\mathbf{h}_t$: use a single-layer GRU (`torch.nn.GRU(input_size=3, hidden_size=d, batch_first=True)`) applied to the sequence of $(r_t, \log V_t, v_t)$ triplets, where $d = 256$ matches the utterance dimension.

5.4 Strategy D: Latent Timestamps via Soft-DTW (Stretch Goal)

Latent clock positions $\tau_1 \leq \tau_2 \leq \cdots \leq \tau_N \in [\tau_s, \tau_e]$ are treated as continuous latent variables. To enforce the monotonicity constraint without hard clipping, parameterise the positions via a cumulative softmax over N logits $\boldsymbol{\ell} \in \mathbb{R}^N$:

$$\tau_i = \tau_s + (\tau_e - \tau_s) \cdot \sum_{j=1}^{i} \frac{\exp(\ell_j)}{\sum_{k=1}^{N} \exp(\ell_k)}. \tag{6}$$

Given positions $\{\tau_i\}$, each utterance i is "spread" onto the minute grid by a Gaussian kernel: $\tilde{\mathbf{u}}_t = \sum_i \exp(-(\tau_i - t)^2/2\sigma^2) \cdot \mathbf{u}_i / Z_t$, where Z_t is the normalisation constant. The predicted volatility sequence $\hat{\mathbf{v}}$ is then obtained from a TCN applied to $(\tilde{\mathbf{u}}_1, \ldots, \tilde{\mathbf{u}}_T)$, and the model is trained end-to-end by minimising:

$$\mathcal{L} = \mathrm{sdtw}_\gamma(\hat{\mathbf{v}},\ \mathbf{v}), \tag{7}$$

where sdtw_γ is the Soft-DTW loss [16] with smoothness parameter γ. Backpropagation through $\mathcal{L}$ with respect to $\boldsymbol{\ell}$ allows the model to learn utterance timing purely from the observed market reaction. This is the most technically ambitious strategy and is most suitable for the AAAI framing.

Implementation note for students. A PyTorch implementation of Soft-DTW is available at `github.com/mblondel/soft-dtw` and also in the `tslearn` library (`pip install tslearn`). The `sdtw_cuda` package provides a fast GPU implementation. Key hyperparameter: $\gamma = 0.1$ (smaller γ approximates hard DTW; larger γ is smoother and easier to optimise). The Gaussian spread kernel bandwidth σ (in minutes) controls how sharply each utterance is pinned to its estimated clock position; start with $\sigma = 2$ minutes and treat as a hyperparameter. The cumulative-softmax parameterisation of τ is implemented as `tau = tau_s + (tau_e - tau_s) * torch.cumsum(F.softmax(logits, dim=0), dim=0)`.

Practical advice: Strategy D requires more careful initialisation and a warm-start procedure. Recommend initialising the logits to produce the uniform prior (all equal) and pre-training for 5 epochs with Strategy A's hard assignment as a pseudo-label before switching to the Soft-DTW loss.

6 Aligned Sequence Prediction Head

Given an assignment matrix $\mathbf{P}$ from any strategy in Section 5, the minute-level fused representation is:

$$\tilde{\mathbf{u}}_t = \sum_{i=1}^{N} P_{it}\, \mathbf{u}_i \in \mathbb{R}^d. \tag{8}$$

A lightweight Temporal Convolutional Network (TCN) with 3 dilated convolutional layers (kernel size 3, dilation factors 1, 2, 4) maps $(\tilde{\mathbf{u}}_1, \ldots, \tilde{\mathbf{u}}_T)$ to the predicted sequence $\hat{\mathbf{v}} \in \mathbb{R}^T$. Pooling the post-call window $[T_e, T_e + 12]$ yields the scalar $\hat{V}_{\text{post}}$ for T2. Pooling the three segment windows yields $\hat{\mathbf{c}}$ for T3.

Implementation note for students. Implement the TCN as three `torch.nn.Conv1d` layers with `dilation=1,2,4`, each followed by ReLU and a residual connection. Input channels $= d = 256$; hidden channels $= 128$; output channels $= 1$ (scalar volatility prediction per minute). The input tensor is shaped (batch, d, T); Conv1d expects channels first. For the prediction head to T2, apply `output[:, :, T_e:T_e+12].sum(dim=-1)` after squeezing the channel dimension. Total trainable parameters in the TCN head: approximately 100K, which trains quickly even on CPU.

7 Ablation Design

Table 1 enumerates all model variants. The design has three orthogonal axes: alignment strategy, input modality, and call scope (full call vs. Q&A only).

The following comparisons test the paper's core claims.

- **Alignment value:** Any strategy (A, B, C, D) vs. the bag-of-utterances baseline with the same modality.
- **Audio value for alignment:** Soft warping (C) with Text+Audio vs. Text only.
- **Q&A focus:** Q&A-only variant vs. full-call variant of Strategy C.
- **Order necessity:** Shuffled utterances placebo vs. Strategy A (same structure, random order). If shuffling degrades performance, sequential order is genuinely used by the alignment.

Table 1: Model variants for the main ablation study.

Variant	Alignment	Input	Scope
Bag-of-utterances	None (mean pool)	Text only	Full call
Bag-of-utterances	None (mean pool)	Text + Audio	Full call
Segment-level (B)	3-anchor	Text + Audio	Full call
Uniform prior (A)	Heuristic	Text + Audio	Full call
Soft warping (C)	Learned OT	Text only	Full call
Soft warping (C)	Learned OT	Text + Audio	Full call
Soft warping (C)	Learned OT	Text + Audio	Q&A only
Soft-DTW (D)	Learned latent τ	Text + Audio	Full call
Shuffled utterances	Strategy A	Text + Audio	Full call
Structured controls only	–	Earnings surprise, size, etc.	–

8 Evaluation Protocol

8.1 Data Splits

All experiments use strict chronological train-validation-test splits. No ECC in the test period may overlap in calendar time with training. Hyperparameters are selected on the validation period; the test set is touched once.

8.2 Primary Metrics

For T1 (intraday sequence), the primary metric is mean Spearman ρ between predicted and observed $\mathbf{v}$ across calls in the test set. For T2 (post-call aggregate volatility), primary metric is Spearman ρ with RMSE as secondary. For T3 (coarse reaction profile), mean absolute error on the $K = 3$ segment profile vector.

8.3 Economic Significance

Sort test-set calls by predicted $\hat{V}_{\text{post}}$. Form a top-minus-bottom decile spread: long calls predicted to have high post-call volatility (e.g., via options straddles) and short calls predicted to have low post-call volatility. Report annualised Sharpe ratio assuming a 60-minute holding window. This connects the statistical results to a financially interpretable interpretation.

8.4 Robustness Checks

(i) **Post-call window variation:** $\Delta \in \{30, 60, 120\}$ minutes.

(ii) **Market-cap stratification:** Small/mid-cap firms are known to exhibit slower price discovery; report results separately.

(iii) **Anchor-time jitter:** Perturb τ_s and τ_e by ± 5 minutes drawn uniformly at random; measure degradation of all alignment strategies.

(iv) **Volume-weight ablation:** Replace $v_t = r_t^2 \cdot \log(1 + V_t)$ with plain r_t^2 and measure whether the volume term improves alignment quality.

9 One-Month Implementation Schedule

Week 1: Data pipeline and baselines. **Day 1–2:** Verify call start/end times for all ECCs in the dataset (check against EDGAR filing timestamps or Refinitiv). Flag calls where the time window is ambiguous or spans market close. **Day 3–4:** Pre-compute and cache all FinBERT text embeddings and Wav2Vec + eGeMAPS audio embeddings; store as HDF5 indexed by (firm, date). **Day 5–7:** Construct T1, T2, T3

labels from the HF data using market-adjusted returns (subtract SPY); implement bag-of-utterances text-only and text+audio baselines; confirm they beat the structured-controls-only logistic regression.

Week 2: Strategies A and B; HF encoder. Implement Strategy A (uniform prior) and the segment-level assignment (Strategy B). Implement the GRU-based HF encoder for $\mathbf{h}_t$. Run the full ablation for T2 and T3 on Strategies A and B. If the segment-level model already improves over the bag baseline for T3, the paper has a publishable minimum result. Decide by Day 14 whether Strategy D (Soft-DTW) is in scope.

Week 3: Strategy C; full ablation. Implement the bilinear cost matrix $\mathbf{S}$ (Eq. 4) and the Sinkhorn OT solver (Eq. 5) using `geomloss`. Implement the cross-modal attention encoder (Eq. 2). Train Strategy C with text-only and text+audio inputs; run the full ablation table including T1. Run the shuffled-utterances placebo, Q&A-only variant, and robustness checks. If Strategy D is in scope, implement the cumulative-softmax parameterisation and Soft-DTW training loop during the second half of this week.

Week 4: Writing and figures. Produce the four required figures: (1) problem diagram showing the two-sequence alignment setting with latent utterance positions; (2) model architecture diagram including cross-modal attention encoder and Sinkhorn module; (3) main ablation table; (4) one representative intraday volume-weighted abnormal volatility curve for a selected call, contrasting Strategy C versus the bag-of-utterances baseline. Write the paper in parallel with any remaining experiments; target a complete draft by Day 26.

10 Scope Constraints and Risk Mitigation

10.1 What Is Out of Scope

The following directions are excluded on grounds of timeline infeasibility: (i) causal attribution of specific utterances to specific price moves, which requires an instrument not available here; (ii) custom speaker diarisation from scratch; (iii) end-to-end fine-tuning of large audio or language models; (iv) minute-level ground-truth alignment claims—the paper's contribution is *learning* alignment, not assuming it.

10.2 Key Risks and Mitigations

Risk 1: Within-call volatility dominated by market noise. Mitigation: use market-adjusted returns and the volume-weighted target (Eq. 1); restrict primary evaluation to after-hours or pre-market calls where idiosyncratic signal is stronger; additionally stratify by earnings surprise quintile and report results for the top quintile separately.

Risk 2: Small effective sample size. If fewer than 500 calls have complete audio coverage, the Sinkhorn OT model may not have enough signal to learn $\mathbf{W}_s$. Mitigation: fall back to Strategy B (segment-level) as the primary result and treat Strategy C as a supplementary experiment.

Risk 3: Sinkhorn OT overfitting. The bilinear matrix $\mathbf{W}_s$ has $d^2 = 65\,536$ parameters. Add L2 regularisation (weight decay 10^{-4}) and monitor validation Spearman ρ for early stopping.

11 Future Directions

Natural extensions include: applying the latent-timestamp framework to analyst days and investor conferences; using SPGISpeech 2.0 [17] (3,780 hours of ECC audio with professional transcripts) for domain-adapted Wav2Vec fine-tuning; extending Strategy D to a variational Bayes formulation over the posterior of latent timestamps; and constructing a public benchmark by deliberately corrupting MAEC [2] timestamp metadata to create a reproducible weakly-aligned evaluation protocol for the broader community.

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